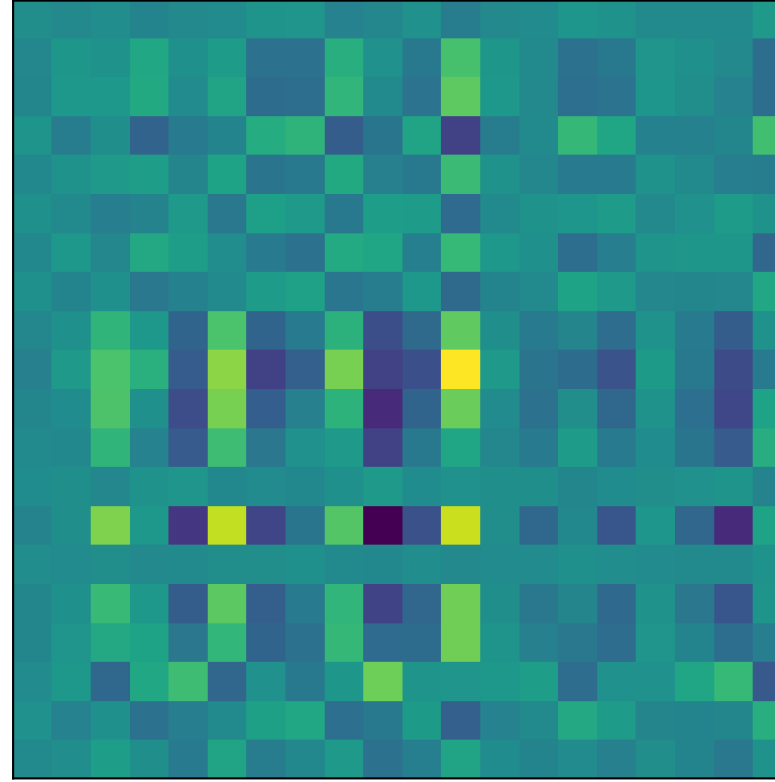
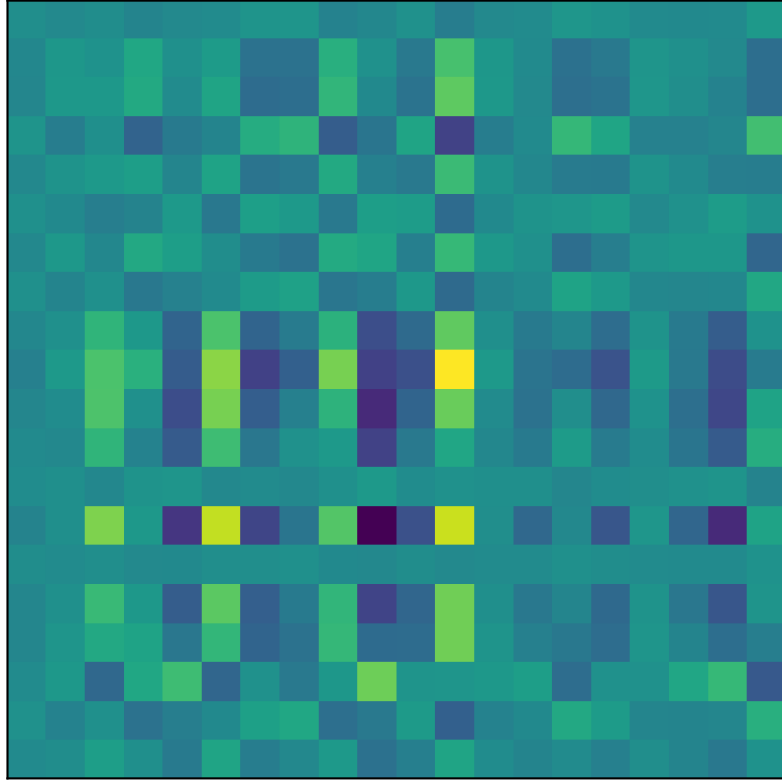
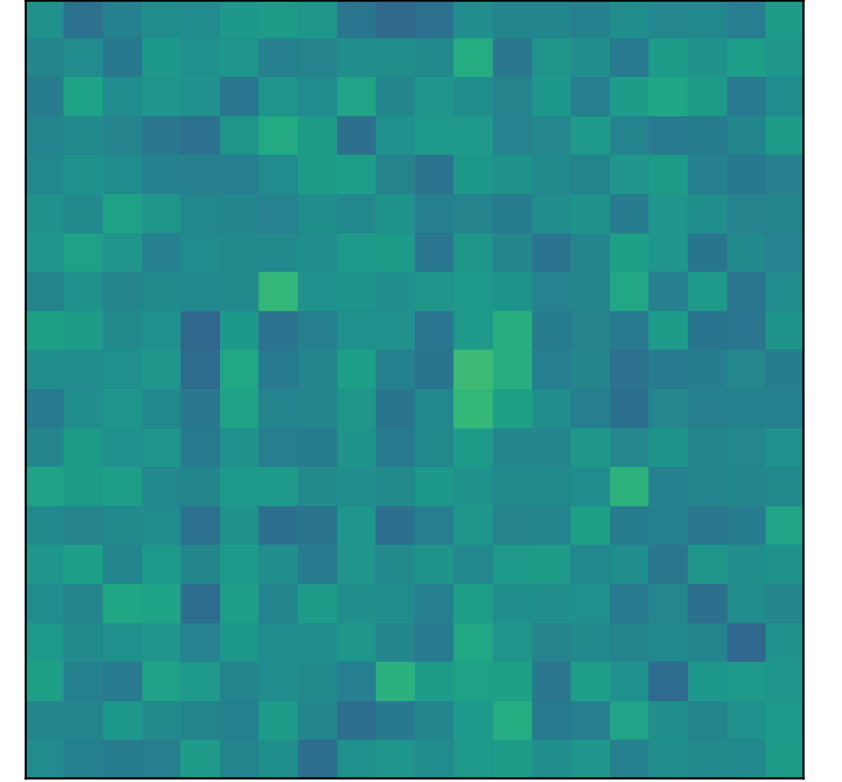


LoRA rank sweep on low-rank shifted benchmark  
 $M = 20, D = M^2 = 400$ , target rank=2, benchmark=rastrigin  
 matched search dims: r=1: d=40, r=2: d=80, r=4: d=160  
 LoRA SVD at r=2,  $d = 2Mr = 80$  ( $A \in \mathbb{R}^{20 \times 2}, B \in \mathbb{R}^{2 \times 20}$ )  
 rel. MSE=2.21e-31

Target  $X_* = A_* B_*$



Best RP at r=2,  $d = 80$  ( $P \in \mathbb{R}^{80 \times 400}$ )  
 rel. MSE=7.59e-01



## Setup

Full space:  $D = M^2 = 400$ .

At each LoRA rank  $r$ , all methods search in the same dimension  $d = 2Mr$ :

LoRA: rank- $r$  factors ( $A, B$ ),  $d = 2Mr$   
 RP: random projection  $P$ ,  $d = 2Mr$   
 RB: random blocking groups,  $d = 2Mr$

Per-rank search dimensions:

r=1 -> d=40  
 r=2 -> d=80  
 r=4 -> d=160

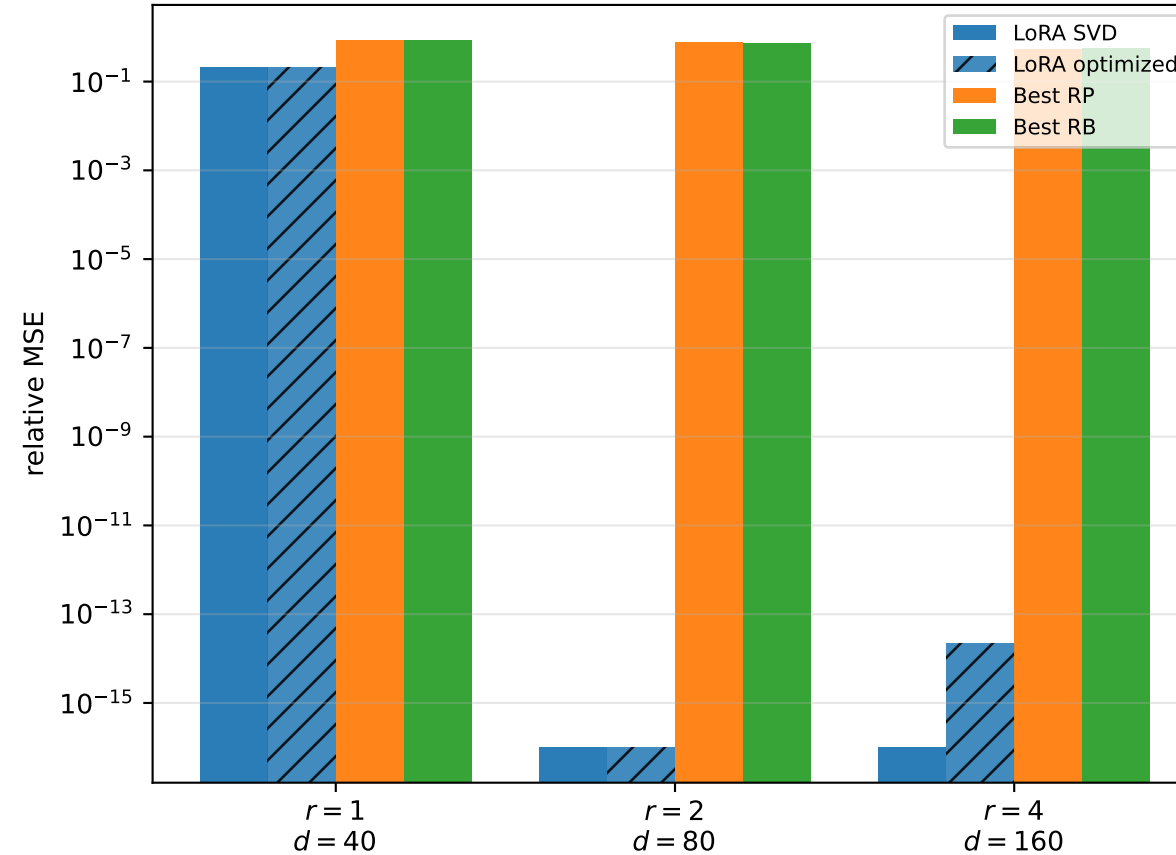
Target rank = 2.

LoRA with  $r \geq$  target rank can represent the optimum exactly (SVD / factor tuning).

RP/RB use fixed random maps; at each rank they only tune  $z$  in the matched  $d$ .

At  $r=2$ , RB uses  $d = 80$  ( $D = 400 \rightarrow \{0, \dots, 79\}$ ).

Reconstruction error vs rank (matched d per method)



Shifted rastrigin vs rank (matched d per method)

